An exact qubit-memory counterexample to an α-Doeblin hockey-stick bound

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Draft of 13 September 2026

Abstract

The complete contraction bound for quantum hockey-stick divergence is known with the positive Doeblin coefficient α_+. A published open question asks whether the stronger coefficient α can replace α_+ when the two inputs have the same reference marginal. We give an exact negative answer. For one use of the qubit amplitude-damping channel with $p = 1$ and $\eta = 1/4$, a noiseless qubit reference, bias $\gamma = 2$, and the entangled pair $|\psi_\pm\rangle = (|00\rangle \pm 10|11\rangle)/\sqrt{101}$, we certify $\alpha = 1/4$, $\alpha_+ = 0$, and

$$E_2\big((\mathrm{id}\otimes\mathcal{N})(\rho)\|(\mathrm{id}\otimes\mathcal{N})(\sigma)\big) - (1-\alpha)E_2(\rho\|\sigma) = \frac{199 + 24\sqrt{41} - 9\sqrt{1489}}{808} > 0.$$

All channel, state, marginal, semidefinite-program, and strict-sign conditions are checked algebraically. The witness disproves the universal α replacement, without claiming a classification or the optimal complete contraction coefficient.

Keywords: quantum channels; strong data processing; hockey-stick divergence; Doeblin coefficient; quantum side information

1 Introduction

A contraction coefficient quantifies how much a channel can preserve the distinguishability of two inputs. Such coefficients strengthen ordinary data processing and enter problems ranging from mixing to noisy quantum hypothesis testing. George, Hirche, Nuradha, and Wilde introduced several quantum analogues of the classical Doeblin coefficient and related them to efficient upper bounds on contraction coefficients [1]. Two variants matter here. The coefficient α optimizes over a Hermitian operator, whereas α_+ restricts that operator to be positive semidefinite. Consequently $\alpha_+ \le \alpha$, and a contraction factor $1-\alpha$ is potentially strictly stronger than $1-\alpha_+$.

For the hockey-stick divergence, the known complete bound uses $1-\alpha_+$. Open Question 1 of Ref. [1] asks whether it remains valid with $1-\alpha$. The question is nontrivial precisely because an optimizer for α need not be positive semidefinite and therefore need not define a physical replacer channel. Follow-up work on conditional contraction retains an α_+ bound [2], while nonlinear hockey-stick inequalities obtained under an additional noise criterion concern a different, non-reference setting [3].

The main contribution is stated and proved with exact arithmetic. A single amplitude-damping channel has $\alpha = 1/4$ but $\alpha_+ = 0$. Two entangled two-qubit inputs with identical reference marginals then violate the proposed $1-\alpha$ bound at $\gamma = 2$. The proof supplies primal and dual semidefinite-program certificates, exact input and output spectra, an attaining measurement effect, and an integer/radical certificate for the strict sign. Because the proposed

statement is universal, this one witness settles it negatively. It does not determine the best replacement factor, describe all violations, or assert failure for every $\gamma > 1$.

2 Definitions and the proposed complete bound

Let $\mathcal{N} : \mathcal{L}(A) \to \mathcal{L}(B)$ be a completely positive, trace-preserving map. We use the unnormalized Choi convention

$$J_{\mathcal{N}} = \sum_{i,j} |i\rangle\langle j|_A \otimes \mathcal{N}(|i\rangle\langle j|), \qquad J_{\mathcal{N}} \succeq 0, \qquad \mathrm{Tr}_B\, J_{\mathcal{N}} = I_A. \tag{1}$$

The two coefficients are

$$\alpha(\mathcal{N}) = \max_{H=H^\dagger} \{\mathrm{Tr}\, H : I_A \otimes H \preceq J_{\mathcal{N}}\}, \tag{2}$$

$$\alpha_+(\mathcal{N}) = \max_{H\succeq 0} \{\mathrm{Tr}\, H : I_A \otimes H \preceq J_{\mathcal{N}}\}. \tag{3}$$

For density operators ρ and σ and $\gamma > 1$, the hockey-stick divergence is

$$E_\gamma(\rho\|\sigma) = \mathrm{Tr}(\rho - \gamma\sigma)_+ = \max_{0\preceq M\preceq I} \mathrm{Tr}\, M(\rho - \gamma\sigma). \tag{4}$$

The established complete inequality says that, for states on $R \otimes A$ with $\rho_R = \sigma_R$,

$$E_\gamma((\mathrm{id}_R \otimes \mathcal{N})(\rho)\|(\mathrm{id}_R \otimes \mathcal{N})(\sigma)) \le [1 - \alpha_+(\mathcal{N})] E_\gamma(\rho\|\sigma) \tag{5}$$

for every $\gamma \ge 1$ [1, Proposition 31]. The proposed strengthening replaces α_+ by α:

$$E_\gamma((\mathrm{id}_R \otimes \mathcal{N})(\rho)\|(\mathrm{id}_R \otimes \mathcal{N})(\sigma)) \overset{?}{\le} [1 - \alpha(\mathcal{N})] E_\gamma(\rho\|\sigma). \tag{6}$$

No ratio is taken in Eq. (6); inputs with zero divergence remain harmless by ordinary data processing.

For priors $1/(1+\gamma)$ on ρ and $\gamma/(1+\gamma)$ on σ, the Helstrom–Holevo binary error probability [4, 5] is

$$P_{\mathrm{err}}(\rho, \sigma) = \frac{1 - E_\gamma(\rho\|\sigma)}{1 + \gamma}. \tag{7}$$

Thus Eq. (6) would lower-bound the post-channel error in a biased discrimination experiment assisted by a noiseless quantum memory.

3 A channel with $\alpha_+ < \alpha$

Consider the generalized amplitude-damping family of Ref. [1, Lemma 10] at $p = 1$ and $\eta = 1/4$. Its two nonzero Kraus operators are

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad K_1 = \begin{pmatrix} 0 & \sqrt{3}/2 \\ 0 & 0 \end{pmatrix}, \qquad \sum_a K_a^\dagger K_a = I. \tag{8}$$

In the ordered Choi basis $(|0_A 0_B\rangle, |0_A 1_B\rangle, |1_A 0_B\rangle, |1_A 1_B\rangle)$,

$$J = \begin{pmatrix} 1 & 0 & 0 & 1/2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3/4 & 0 \\ 1/2 & 0 & 0 & 1/4 \end{pmatrix}. \tag{9}$$

It is positive semidefinite, with spectrum $\{0, 0, 3/4, 5/4\}$, and $\mathrm{Tr}_B\, J = I_A$.

Lemma 1. *For the channel in Eq. (8),*

$$\alpha(\mathcal{N}) = \frac{1}{4}, \qquad \alpha_+(\mathcal{N}) = 0. \tag{10}$$

Proof. For α, choose

$$H = \mathrm{diag}(1/2, -1/4), \qquad Y = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}. \tag{11}$$

The primal slack $J - I \otimes H$ has spectrum $\{0, 1/4, 1/4, 1\}$, so H is feasible and gives $\mathrm{Tr}\, H = 1/4$. Moreover $Y \succeq 0$, $\mathrm{Tr}_A\, Y = I_B$, and $\mathrm{Tr}(JY) = 1/4$. The matching primal and dual objectives certify $\alpha = 1/4$. This also reproduces the family formula $\alpha(\mathcal{N}_{p,\eta}) = (1 - \sqrt{\eta})^2$ at the selected point.

For α_+, the zero matrix is primal feasible. Let

$$|\kappa\rangle = |00\rangle - 2|11\rangle, \qquad Y_+ = |\kappa\rangle\langle\kappa|. \tag{12}$$

Then $J|\kappa\rangle = 0$ and $\mathrm{Tr}_A\, Y_+ = \mathrm{diag}(1, 4) \succeq I$. For every positive primal-feasible H_+,

$$\mathrm{Tr}\, H_+ \leq \mathrm{Tr}[(\mathrm{Tr}_A\, Y_+)H_+] = \mathrm{Tr}[Y_+(I \otimes H_+)] \leq \mathrm{Tr}(Y_+ J) = 0. \tag{13}$$

Together with the zero primal point, this proves $\alpha_+ = 0$. □

4 Exact counterexample

All state matrices below use the R-major order $(|0_R 0_A\rangle, |0_R 1_A\rangle, |1_R 0_A\rangle, |1_R 1_A\rangle)$ before the channel, and the corresponding R, B order afterwards.

Theorem 1. *Equation (6) is false even when $A = B = R = \mathbb{C}^2$. A counterexample is given by the channel in Eq. (8), $\gamma = 2$, and*

$$|\psi_\pm\rangle = \frac{|00\rangle \pm 10|11\rangle}{\sqrt{101}}, \qquad \rho = |\psi_+\rangle\langle\psi_+|, \quad \sigma = |\psi_-\rangle\langle\psi_-|. \tag{14}$$

Proof. The two operators in Eq. (14) are rank-one density operators and have the identical reference marginal

$$\rho_R = \sigma_R = \mathrm{diag}(1/101, 100/101). \tag{15}$$

The eigenvalues of either partial transpose include $-10/101$, so the inputs are entangled and cannot reduce to a common classical-reference construction.

The only nonzero block of $\rho - 2\sigma$, on $\mathrm{span}\{|00\rangle, |11\rangle\}$, is

$$\frac{1}{101}\begin{pmatrix} -1 & 30 \\ 30 & -100 \end{pmatrix}. \tag{16}$$

Its trace and determinant are -1 and $-800/101^2$, respectively, so it has exactly one positive eigenvalue. Direct diagonalization gives

$$E_2(\rho\|\sigma) = \frac{3\sqrt{1489} - 101}{202}. \tag{17}$$

The channel action is

$$\mathcal{N}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a + 3d/4 & b/2 \\ c/2 & d/4 \end{pmatrix}. \tag{18}$$

Applying it to the A subsystem gives

$$\rho' = \frac{1}{101}\begin{pmatrix} 1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 75 & 0 \\ 5 & 0 & 0 & 25 \end{pmatrix}, \qquad \sigma' = \frac{1}{101}\begin{pmatrix} 1 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 75 & 0 \\ -5 & 0 & 0 & 25 \end{pmatrix}. \tag{19}$$

The active block of $\rho' - 2\sigma'$ is $101^{-1}\left(\begin{smallmatrix} -1 & 15 \\ 15 & -25 \end{smallmatrix}\right)$; the remaining eigenvalues are 0 and $-75/101$. Hence

$$E_2(\rho' \| \sigma') = \frac{3\sqrt{41} - 13}{101}. \tag{20}$$

For completeness, the positive eigenspace is attained by the effect whose active two-dimensional block is

$$M_{\mathrm{act}} = \frac{1}{2}\left[I + \frac{5\sigma_x + 4\sigma_z}{\sqrt{41}}\right]. \tag{21}$$

Indeed $M_{\mathrm{act}}^2 = M_{\mathrm{act}}$ because $5^2 + 4^2 = 41$; embedding it by zero on the other two basis vectors gives $0 \preceq M \preceq I$ and reproduces Eq. (20) through Eq. (4).

Using Eq. (10), the excess over the proposed right-hand side is

$$\Delta = \frac{3\sqrt{41} - 13}{101} - \frac{3}{4}\frac{3\sqrt{1489} - 101}{202} = \frac{199 + 24\sqrt{41} - 9\sqrt{1489}}{808}. \tag{22}$$

It remains to certify strictness without a floating-point comparison. Both sides of $199 + 24\sqrt{41} > 9\sqrt{1489}$ are positive, and their squared difference is

$$-57392 + 9552\sqrt{41} > -57392 + 9552\frac{32}{5} = \frac{18704}{5} > 0, \tag{23}$$

where $\sqrt{41} > 32/5$ follows by squaring. Therefore $\Delta > 0$, which contradicts Eq. (6). □

5 Discrimination meaning and comparison with known bounds

Figure 1 displays decimal renderings of the exact values only for visual comparison. Ordinary data processing remains intact: $E_2(\rho' \| \sigma') \approx 0.061479$ is below the input value 0.073083. The known α_+ inequality (5) also remains valid, since $\alpha_+ = 0$. What fails is the stronger proposed ceiling $(1 - \alpha)E_2 \approx 0.054812$.

For priors $1/3$ on ρ and $2/3$ on σ, Eq. (7) gives an actual post-channel error of approximately 0.312840. If Eq. (6) held, it would require an error of at least 0.315063. The proposed guarantee therefore overstates the unavoidable loss of biased discrimination power by exactly $\Delta/3$ for this memory-assisted experiment.

The role of the quantum reference is essential to the example. Without a reference system, the ordinary hockey-stick contraction coefficient is bounded by the trace-distance coefficient and hence by $1 - \alpha$ [1, Section 6.1]; a common classical reference basis likewise reduces the problem blockwise [2, Appendix A]. The negative partial-transpose eigenvalue in Eq. (15) rules out those reductions. The strict gap $\alpha_+ < \alpha$ also rules out the region already covered by equality of the two Doeblin coefficients.

Among the directly related works checked through 13 September 2026, the conditional framework of Ref. [2] continues to use α_+ and explicitly discusses the obstruction posed by a quantum reference at general γ. The nonlinear bounds of Ref. [3] assume an additional channel-noise condition and optimize without the same-reference complete setting. These comparisons are bounded rather than an exhaustive priority search; the result is therefore presented without a priority claim.

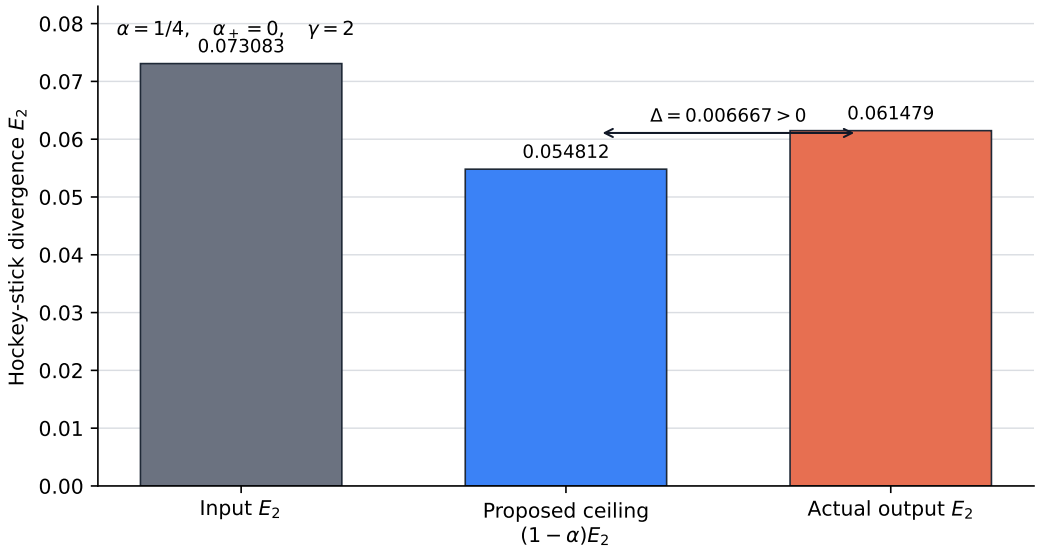


Figure 1: Exact witness values, plotted in decimal form. The actual output divergence exceeds the proposed α-based ceiling by the exact positive quantity Δ in Eq. (22). The plot is generated directly from the symbolic expressions; no numerical fit or optimization enters the proof.

6 Scope and reproducibility

The theorem uses one exact, finite-dimensional channel instance and one exact pair of input states. It is sufficient to disprove both the qubit-universal statement and the arbitrary-dimensional universal proposal that contains it. It does not characterize the optimal complete contraction coefficient, all violating channels or inputs, the full range of γ, or robustness to experimental imperfections. No perturbative, asymptotic, continuum, or unit-dependent statement is involved.

The ancillary package contains two independent exact implementations. One constructs the Kraus action, Choi matrix, semidefinite-program certificates, states, spectra, and effect using symbolic rational and radical arithmetic. The other begins from the rational state matrices and the block action in Eq. (18), imports no result from the first checker, and repeats the decisive characteristic-polynomial and sign checks. A separate script generates Fig. 1 and its tabulated values from the exact expressions. The environment and commands are specified in `REPRODUCIBILITY.md`.

Data and code availability

All witness matrices are included in the ancillary JSON data, and all exact verification and figure-generation scripts are included in the local source package. No downloaded third-party paper is part of that package. A public archival location and license for the code must be selected by the prospective author before dissemination or submission.

Author contributions and AI-use disclosure

This is a provisional local draft for author review. Xiaojian Zeng is the prospective corresponding author. OpenAI Codex language models were used for idea development, the exact calculations and code, a bounded bibliographic check, figure production, and drafting and editing the manuscript. The models are not authors. Before any dissemination or submission,

the prospective author must independently inspect the sources, calculations, code, claims, text, and rights; make any needed corrections; and explicitly accept full responsibility. That human review and approval are not represented as complete here.

Competing interests

The prospective author has not yet supplied a competing-interest declaration; it must be completed before submission.

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